

Effect of Ski Binding Parameters on Knee Biomechanics: A Three-Dimensional Computational Study

NANCY ST-ONGE¹, YAN CHEVALIER¹, NICOLA HAGEMEISTER^{1,2},
MAXIME VAN DE PUTTE¹, and JACQUES DE GUISE^{1,2}

¹Laboratory in Imaging and Orthopedic Research, Research Center of the University of Montreal Hospital Center, Notre-Dame Hospital, Montreal, CANADA; and ²Ecole de Technologie Supérieure (ETS), University of Quebec in Montreal (UQAM), Montreal, CANADA

ABSTRACT

ST-ONGE, N., Y. CHEVALIER, N. HAGEMEISTER, M. VAN DE PUTTE, and J. DE GUISE. Effect of Ski Binding Parameters on Knee Biomechanics: A Three-Dimensional Computational Study. *Med. Sci. Sports Exerc.*, Vol. 36, No. 7, pp. 1218–1225, 2004.

Introduction: Downhill skiing is a relatively safe sport, but many potentially avoidable injuries do occur. Whereas tibia and ankle injuries have been declining, severe knee sprains usually involving the anterior cruciate ligament (ACL) have increased from the 1970s to the 1990s. The goal of the present study was to evaluate the effect of the position of the binding pivot point and binding release characteristics on ACL strain during a phantom-foot fall. **Methods:** We computed ACL strain using a biomechanical computer knee model to simulate the phantom-foot ACL-injury mechanism. This mechanism, which is one of the most common mechanisms of ACL injury in downhill skiing, occurs when the weight of the skier is on the inner edge of the ski during a backward fall, resulting in a sharp uncontrolled inward turn of the ski. **Results:** The model predicts, that under simulated phantom-foot conditions, a binding with fast-release characteristics with a pivot positioned in front of the center of the boot produces less strain on the ACL. Current bindings have their pivot point approximately at the center of the heel radius. A pivot positioned at the back of the binding is more effective for sensing loads that occur at the tip of the ski. However, it is less effective for sensing loads that occur at the tail of the ski and, therefore, offers less protection during a phantom-foot fall. **Conclusion:** A binding with two pivot points, one positioned in front and the other at the back, could sense twist loads applied to the ski both at the front and at the back, and might, therefore, be a solution to reduce the occurrence of ACL injuries. **Key Words:** SKI INJURY, PHANTOM-FOOT, ACL STRAIN, BINDING RELEASE CHARACTERISTICS, PIVOT POINT

Downhill skiing is a relatively safe sport. Nevertheless, bindings allowing rigid fixation of the boot with the ski and long skis acting as levers through which forces can be transmitted to the lower extremity may expose skiers to many different types of lower-extremity injuries. The overall injury rate was estimated to be between 1 and 6 per 1000 skier-days in different studies from the 1970s to the 1990s (5,16,20,27,28). However, most authors

agree that the incidence of injury has declined by approximately 40–50% over the years (5,17,18).

Johnson et al. (17) observed that most of the decline in skiing injuries could be attributed to a 60% reduction in the incidence of lower-extremity injuries. In fact, lower-extremity injuries, which account for 40–60% of all injuries (16,20), have been decreasing over the last 20 yr at Sugarbush, Vermont (16–18). Whereas tibia and ankle injuries have been decreasing by about 70–90% (5,17), severe knee sprains usually involving the anterior cruciate ligament (ACL) have been increasing by as much as 170–280% from the 1970s to the 1990s (5,17,18). In the late 1980s, ACL rupture was found to represent 2.4% of all injuries (20). However, from 1972 to 1994, grade-III sprain of the ACL was the most common injury in adult skiers, accounting for 18.4% of all injuries (5). According to J. E. Shealy (personal communication, 2002), the increase in ACL injuries occurred after the decline in tibial shaft fractures and is not related to binding system improvements that caused the

Address for correspondence: Professor Jacques de Guise, LIO, Notre-Dame Hospital, Pav. J.A. de Seve, 1560 Sherbrooke East, Y-1614, Montreal (Quebec) Canada H2L 4M1; E-mail: jacques.deguise@etsmtl.ca.

Submitted for publication April 2003.

Accepted for publication February 2004.

0195-9131/04/3607-1218

MEDICINE & SCIENCE IN SPORTS & EXERCISE®

Copyright © 2004 by the American College of Sports Medicine

DOI: 10.1249/01.MSS.0000132375.00721.7A

salutary effect on tibial shaft fractures. Shealy believes that other factors that made their appearance in the 1970s are responsible for the increase in ACL sprains. These changes include new boot and ski designs but also the popularity of groomed snow surfaces. Interestingly, although Johnson et al. (16) observed no statistically significant reduction in the lower-extremity nonequipment-related injury rate, they found a significant decline in lower-extremity equipment-related injuries. In fact, it is widely recognized that the decrease in the rate of lower-extremity injuries may be due to improvements in ski-boot-binding systems. For example, taller and stiffer boots give greater protection to the foot and ankle. Also, bindings, which are more effective in terms of releasing capability, offer better protection against tibial fractures caused by bending and twisting mechanisms.

Five major mechanisms causing ACL injury during downhill skiing have been described: the boot-induced anterior drawer mechanism (15), valgus-external rotation (9,23), a combination of internal rotation and extension (23), a forceful quadriceps contraction (6), and the phantom-foot injury mechanism (15). The phantom-foot injury mechanism is imminent when the following six elements occur: uphill arm back, skier off-balance to the rear, hips below the knees, uphill ski unweighted, weight on the inside edge of downhill ski tail, and upper body generally facing downhill ski. The modern parabolic alpine ski is designed to turn more easily, at different radii, when the weight is put on its edge. If the weight is on the inner edge of the ski during a backward fall, a sharp uncontrolled inward turn of the ski can be induced, leading to an isolated ACL rupture (9). The phantom-foot is considered to be the most common or one of the most common mechanisms of ACL injury in downhill skiing (5,8; J. E. Shealy, personal communication, 2002).

Current bindings have three release modes: upward at the heel, upward at the toe, and lateral at the toes. In the latter case, the settings are adjusted such that the toe cup will release the boot toe when the lateral force at the boot toe exceeds a certain level. The boot then pivots about an axis near the boot heel. The position of the pivot as well as the binding release characteristics determine the combination of forces and moments needed to release the boot from the binding and could have an effect on ACL strain during skiing. *In vitro* and *in vivo* experimental protocols can be of tremendous help in understanding knee injury mechanisms. However, they also face different methodological problems that can be alleviated by computer modeling.

In the present study, we developed a biomechanical computer knee model to study ACL strain during a ski injury mechanism. We chose to reproduce the phantom-foot injury mechanism with our model because it is one of the most common mechanisms of ACL injury in downhill skiing and is also generated by a twist load. The main goal of the present study was to evaluate the effect of the position of the pivot point on ACL strain and thus on the chances of tearing the ligament. The second goal was to evaluate the effect of binding release characteristics on ACL strain.

METHODS

A biomechanical model was assembled to represent a dynamic system consisting of a knee joint, developed in a previous study (Chevalier, Y. Creation of a three-dimensional biomechanical model of the knee for the preoperative evaluation of ligament reconstructions. M.A.Sc. Thesis, Ecole Polytechnique of Montreal, Montreal, Canada, 2002, pp. 1–206), a ski, a boot, and a binding. The full model was implemented on a PC workstation with commercial software (MADYMO, TNO Automotive), which solves equations of motion according to a first-order, modified Euler integration method with a time step of 0.001 s. The model was shown to reproduce *in vitro* extension three-dimensional (3D) kinematics (Chevalier, Y., M.A.Sc. Thesis) taken from a study by Hagemester et al. (13).

Knee model. Segmentation and polygon-generation techniques, performed with commercial software (Slice-O-Matic, Tomovision, Montreal, Canada), were used to reconstruct 3D rigid surface envelopes of the femur, tibia, and fibula, from computer-assisted tomographic images of a cadaver knee. For the modeled bone structures, physical properties of mass and inertia were taken from Piazza and Delp (26) and are shown in Table 1. The z-axis coincides with the anatomical axis defined by the diaphysis of the tibia. The menisci were also reconstructed from magnetic resonance images. These structures were assumed to be rigid, but penetration was allowed at their surfaces. Stress and surface penetration are related by elastic contact formulation, which was adapted from a linear stress-strain formulation and material properties proposed by Blankevoort and Huiskes (4): Young's modulus of 5 MPa and Poisson's ratio of 0.45. A constant damping coefficient of $4.34 \times 10^{-7} \text{ N} \cdot \text{s} \cdot \text{m}^{-1}$ was chosen to relate damping stress and penetration velocity. This value was chosen to qualitatively reproduce *in vitro* extension 3D kinematics (13). Friction between tibial and femoral surfaces was defined with a friction coefficient of 0.008, based on the literature (24). The material properties of the menisci were described with a Young's modulus of 4 MPa, and a Poisson ratio set at 0.45, based on work by Bendjaballah et al. (1). Values of 0.001 and 0.08 were used for femoro-meniscal and meniscotibial friction coefficients, respectively, based on optimization. Six parallel nonlinear spring-and-damper elements (Kelvin elements) were taken to model each ligament (ACL, posterior cruciate ligament, medial collateral ligament, lateral collateral ligament). Force-length behavior of the modeled ligaments was adapted from Blankevoort and Huiskes (4), with mechanical properties defined by Marinozzi et al. (22). A damping coefficient for each Kelvin element was set

TABLE 1. Physical properties of moving model components.

	Mass (kg)	Inertia Tensor ($\text{kg} \cdot \text{m}^{-2}$)					
	m	I_{xx}	I_{yy}	I_{zz}	I_{xy}	I_{xz}	I_{yz}
Tibia and fibula	3.75	0.065	0.065	0.019	0	0	0
Boot	3.75	0.0375	0.0375	0.019	0	0	0
Binding	1.78	0.03125	0.00313	0.02924	0	0	0.00434
Ski	1.60	0.28527	0.0008	0.28583	0.00033	0	0

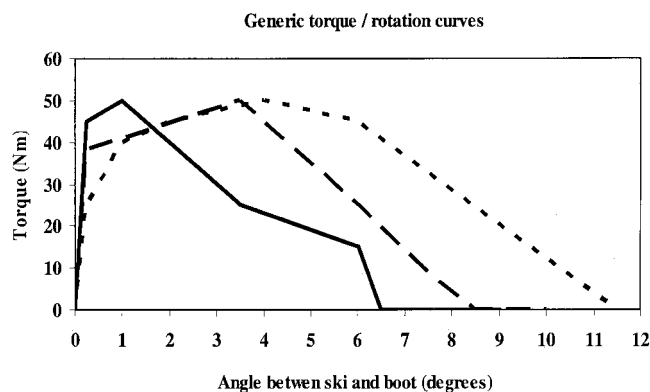


FIGURE 1—Binding release characteristics. Phantom-foot kinematics were simulated using three release characteristics: fast release (solid line), slow release (dotted line), or intermediate release (dashed line).

at $390.5 \text{ N} \cdot \text{s} \cdot \text{m}^{-1}$ to reproduce *in vitro* extension 3D kinematics (13).

Ski, boot, and binding components. Ski, binding, and boot components were modeled as rigid bodies. Their physical properties are presented in Table I. Boot-foot inertia was calculated from the inertia tensor of a cylinder with length 0.30 m and radius 0.10 m. Ski and binding inertia tensors were acquired from measurements on commercially available ski equipment. Because this study was geared to competitive ski racers, the tibia was attached to the boot rigidly, forming an angle of 20° with the boot sole in the sagittal plane. The release action of the binding was modeled on a 2D pivot joint, with a rotational axis perpendicular to the ski. Stiffness of this pivot joint was adjusted to simulate the release characteristics of the binding. Three typical release characteristics were tested: fast release, slow release, or intermediate release (Fig. 1). Peak loads of 50 N·m were chosen. This is the recommended release setting for a 50 percentile adult male.

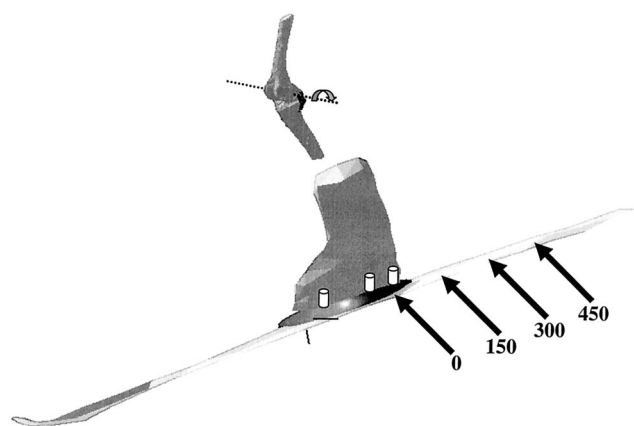


FIGURE 2—Pivot points and applied load positions. Phantom-foot kinematics were simulated by applying external loads (arrows) of 600-N magnitude on the medial edge of the ski 150, 300, or 450 mm behind the projection of the tibial axis and using three pivot positions. We also applied a 600-N load at the projection of the tibial axis even though it does not reproduce the phantom-foot injury mechanism. Pivots (white cylinders) were positioned relative to the center of the boot: 63 mm in front, 63 mm in the back, or 135 mm in the back.

Simulation of phantom-foot injury. To simulate the phantom-foot injury mechanism, external loads were applied on the medial edge of the ski (Fig. 2). Loads of a magnitude of 600 N were applied 150, 300, or 450 mm behind the projection of the tibial axis while the femur was rotated at a rate of $4 \text{ rad} \cdot \text{s}^{-1}$. A 600-N load was also applied at the projection of the tibial axis even though it might not reproduce the phantom-foot mechanism. We still chose to apply such a load because it is of interest to know how the binding reacts to loads applied near the pivot point. The knee flexion which takes place during the phantom-foot mechanism was simulated by imposing a flexion movement on the femur about a fixed axis passing through both femoral condyles. The initial position of the knee was about 45° of flexion, 0° of abduction, and 10° of external rotation which corresponds to a position that was defined from experimental data from a cadaveric experiment when no abduction/adduction or internal/external rotation moments were applied (13).

Coordinate systems were defined on both the femur and tibia by identifying anatomical landmarks on the femur (center of the lateral and medial condyles and center of the femoral shaft) and on the tibia (center of circles representing the medial and lateral tibial plateaus and center of the tibial shaft). The knee joint coordinate system was constructed in the following way: The flexion/extension axis links the lateral and medial points on the femur, the tibial rotation axis is perpendicular to the axis joining the medial and lateral tibial plateaus and lies on the plane defined by the medial and lateral plateaus and the center of the tibial shaft, and the ab/adduction axis is obtained by the vectorial cross-product of the two other axes. This enabled us to compute 3D kinematics according to the method of Grood and Suntay (11). We investigated flexion/extension, abduction/adduction, internal/external tibial rotation, as well as antero-posterior displacement. ACL length was computed from the distance between the femoral and tibial attachment sites that were digitized on the cadaveric specimen. For the computation of ligament strain, the reference length of the ACL was defined when the knee was in 45° of flexion.

Pivot point position. Three pivot positions were tested. They were positioned relative to the center of the boot: 63 mm in front, 63 mm in the back, or 135 mm in the back (Fig. 2). The 135-mm point is approximately half way between the center of curvature of the heel and the extreme heel end of a 300-mm boot sole. The best case pivot point would be located at the center of curvature (116 mm back from the center of a 300-mm boot sole) because the boot would tend to rotate about the heel center if there were no friction. However, it would tend to rotate about one of the contact points between the boot and binding heel cup if friction were very high. Therefore, some point in between (135 mm in the back) seems most appropriate to represent the typical pivot point. The two other points (63 mm in front, 63 mm in the back) were chosen because they are common to a new binding functioning with two effective pivot points.

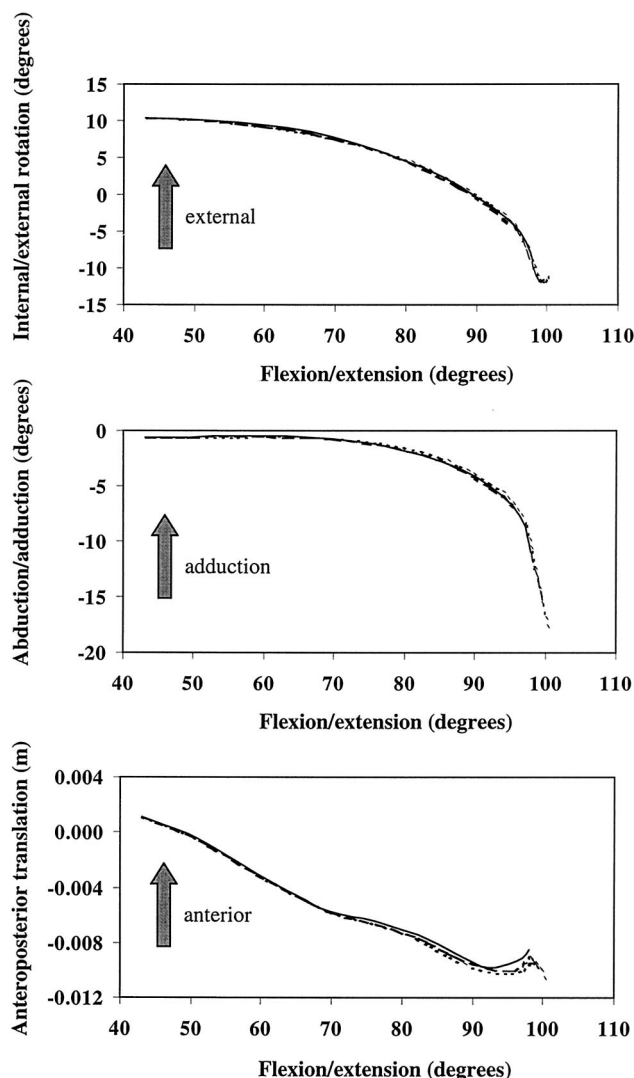


FIGURE 3—Phantom-foot kinematics. When a 600-N load was applied on the medial side of the ski 150 mm behind the projection of the tibial axis using intermediate release characteristics, the knee went into internal rotation and abduction as it flexed. Also, the tibia translated posteriorly on the femur. The pivot was positioned 63 mm in front (solid line), 63 mm in the back (dashed line), or 135 mm in the back (dotted line) relative to the center of the boot.

RESULTS

Phantom-foot kinematics. When a load was applied on the medial side of the ski behind the projection of the tibial axis, the knee went into internal rotation and abduction as it flexed, which is typical of the phantom-foot mechanism. Also, the tibia translated posteriorly on the femur. As an example, Figure 3 shows the kinematics obtained when a 600-N load was applied 150 mm behind the projection of the tibial axis, using intermediate release characteristics for the three pivot points. Figures 4 and 5 show the range of internal rotation and abduction through which the knee moved when a 600-N load was applied 150, 300, or 450 mm behind the projection of the tibial axis. The results are given for a pivot point positioned 63 mm in front, 63 mm in the back, or 135 mm in the back of the center of the boot for fast, slow, and intermediate release characteristics. When a

load was applied at the projection of the tibial axis, the knee first went into internal rotation and abduction as it flexed. However, for some pivot positions, the knee movement later changed direction and went into external rotation and adduction (Fig. 6), which is not typical of the phantom-foot mechanism.

Simulated ACL strain. When a load was applied on the medial side of the ski as a flexion movement was imposed on the femur, ACL strain generally increased as the movement progressed (Fig. 7). For the fast release characteristic, especially when the load was applied further back from the projection of the tibial axis, ligament strain did not have time to increase before the binding released (not shown). Figure 8 shows the maximal ACL strain produced when a 600-N load was applied at the projection of the tibial axis as well as 150, 300, or 450 mm behind that point. The results are presented for a pivot point positioned 63 mm in front, 63 mm in the back, or 135 mm in the back of the center of the boot for fast, slow, or intermediate release characteristics. Ligament strain was always less when the pivot point was positioned 63 mm in front of the center of the boot. Also, ligament strain was usually less when the pivot was positioned 63 mm behind than when it was

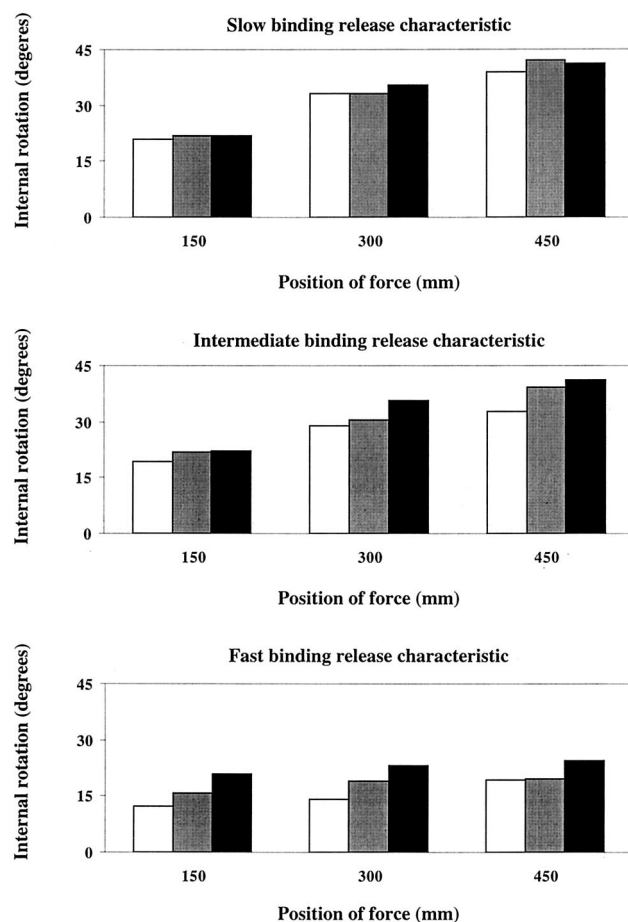


FIGURE 4—Range of internal rotation during phantom-foot simulation. A 600-N load was applied 150, 300, or 450 mm behind the projection of the tibial axis. The pivot was positioned 63 mm in front (white bar), 63 mm in the back (gray bar), or 135 mm in the back (black bar) relative to the center of the boot.

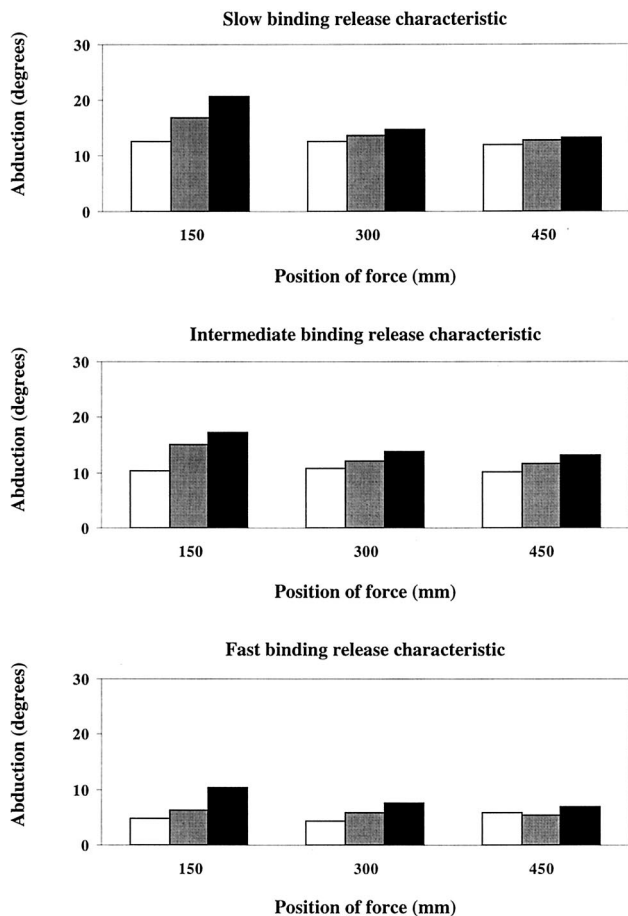


FIGURE 5—Range of abduction during phantom-foot simulation. A 600-N load was applied 150, 300, or 450 mm behind the projection of the tibial axis. The pivot was positioned 63 mm in front (white bar), 63 mm in the back (gray bar), or 135 mm in the back (black bar) relative to the center of the boot.

positioned 135 mm behind the center of the boot. Note that ligament strain rarely reached 35% which was found to be the strain at maximal load with a loading rate of 500 mm·min⁻¹ (19).

The data presented in Figure 8 were rearranged to compare ACL strains generated when simulating a same pivot position but with different characteristics (Fig. 9). For all three pivot points, ACL strain was less when the fast binding characteristic was simulated, whereas the results were similar for the slow and intermediate characteristics. When the load was applied at the projection of the tibial axis for a pivot point positioned 135 mm behind the center of the boot, the ACL was highly strained, independently of the binding release characteristic being simulated.

DISCUSSION

ACL ski injury epidemiology. The decrease in the rate of lower-extremity injuries that has been observed over the years is probably due to improvements in ski-boot-binding systems. However, whereas tibia and ankle injuries have been falling, severe knee sprains usually involving the ACL have been increasing from the 1970s to the 1990s.

As far as we can now tell, current toe and heel conventional systems are excellent at preventing ankle and tibial injuries, but, according to Natri et al. (25), are “virtually worthless against serious knee injuries.” To the best of our knowledge, the loads necessary to injure the ACL during the phantom-foot injury mechanism have never been measured. It has been suggested by Natri et al. (25) that the unique combination of loads applied to the knee under such conditions may produce injury at a relatively low level. According to Ettlinger and Johnson (7), current binding designs are unable to provide adequate protection for the knee because the load required to injure the hyperflexed knee is less than the load required for normal skiing maneuvers.

Computer model. Test methods defined by the ASTM and ISO have been used to evaluate ski bindings. These tests give information on the torsion and bending loads that would be transmitted to a skier’s lower leg during various types of falls. However, they do not provide much insight into the loads transmitted to the skier’s knee during such falls.

To better understand ACL injury mechanisms and effect, researchers have developed *in vitro* and *in vivo* experimental protocols permitting the study of knee biomechanics. Although these studies have proven to be of tremendous help in understanding knee injury causes and effects, they also

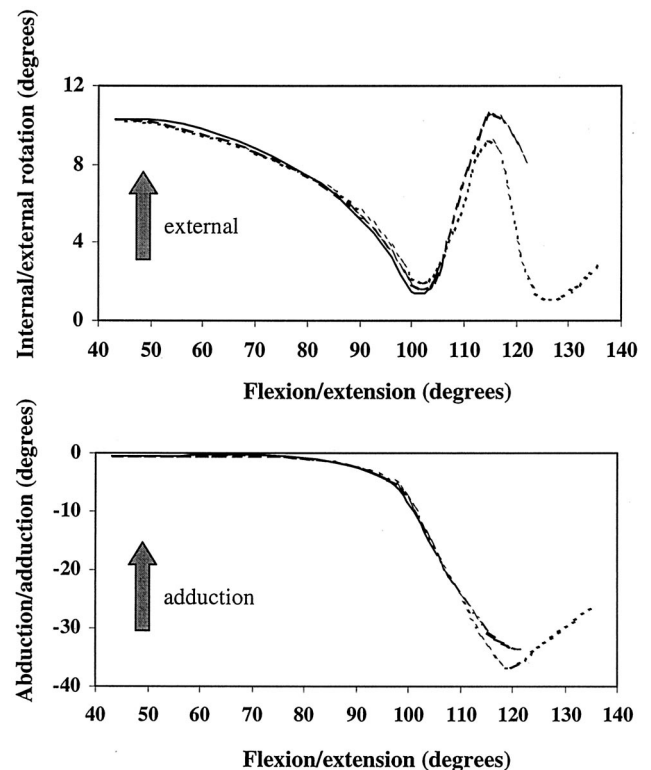


FIGURE 6—Kinematics generated by applying a 600-N load on the medial side of the ski at the projection of the tibial axis using intermediate release characteristics. The knee first went into internal rotation and abduction as it flexed. However, for some pivot positions, the knee movement later changed direction and went into external rotation and adduction, which is not typical of the phantom-foot mechanism. The pivot was positioned 63 mm in front (solid line), 63 mm in the back (dashed line), or 135 mm in the back (dotted line) relative to the center of the boot.

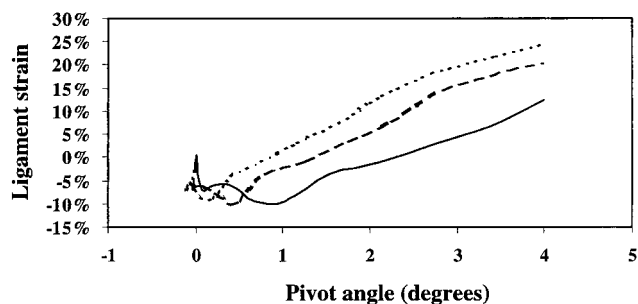


FIGURE 7—Ligament strain versus pivot angle (angle between ski and boot). Phantom-foot kinematics were generated with a 600-N load applied 150 mm behind the projection of the tibial axis using intermediate release characteristics. The pivot was positioned 63 mm in front (solid line), 63 mm in the back (dashed line), or 135 mm in the back (dotted line) relative to the center of the boot.

face different methodological problems that can be alleviated by computer modeling.

In vitro studies are usually done through test rigs that simulate joint loads on cadaveric specimens. Although they have been, and still are, very helpful in understanding knee biomechanics, test rigs that are designed to simulate life-like situations on a cadaveric knee are complicated and tend to

be unstable due to the multiple forces applied at different levels and locations on the specimen (12). Also, it is impossible to simulate realistic loads because the application of such loads through tendons makes the latter slip or rupture (14). Limitations of *in vitro* studies also include the variability of soft tissue properties among specimens and the number of specimens needed to simulate a large number of injury scenarios (3). It has been reported that maximal tensile strength of the soft tissues of different specimens could vary by over 100% (3). Specimen integrity is also an issue. Because a sprained ligament will not return to its initial length (21), specimens that are suspected to have surpassed their structural capabilities have to be discarded. Simulating the phantom-foot fall *in vitro* would have demanded a very complicated test rig and a very large number of specimens, because we wanted to study ACL strain during a phantom-foot fall with many pivot points and release characteristics.

For this study, *in vivo* evaluation of strain being imposed on the ACL during the phantom-foot fall would have been impossible. Although many researchers have measured *in vivo* ACL strain during different types of functional exercises using strain gauges implanted on the subject's ACL (2,10), the highly invasive nature of this procedure renders it inappropriate for an evaluation done outside of the operating room. Also, realistic on-snow replication of the phantom-foot mechanism would incur too large of a risk of rupturing the subject's ACL.

In the present study, we computed ACL strain using a biomechanical computer knee model to simulate the phantom-foot ACL injury mechanism. We used computer modeling to better control the uncertainties and overcome methodological difficulties associated with *in vitro* and *in vivo* investigations. A computer model is an interesting way of overcoming the difficulties associated with *in vitro* and *in vivo* experimentation. It does permit the evaluation of a large number of situations, and therefore the comparison of many pivot points and binding characteristics without taking the risk of injuring subjects.

Effect of binding release characteristics and pivot point position. Current bindings have their best-case pivot point approximately at the center of the heel radius. However, friction moves the pivot point further back. Therefore, the typical pivot point is somewhere in between the center of the heel radius and the back of the boot. Because the pivot point is behind the projected tibial axis, these bindings are therefore more effective at sensing loads that are applied at the front of the ski than those applied at the back. During the phantom-foot mechanism, the weight of the skier is on the inside edge of the downhill ski tail. The force that makes the ski carve inside should, therefore, act on the back of the ski. The goal of the present study was to evaluate the effect of the position of the pivot point on ACL strain and thus on the chances of tearing the ligament. When a 600-N load was applied at the projection of the tibial axis, as well as 150, 300, or 450 mm behind that point, ligament strain was always less when the pivot point was positioned 63 mm in front of the center of the boot. For

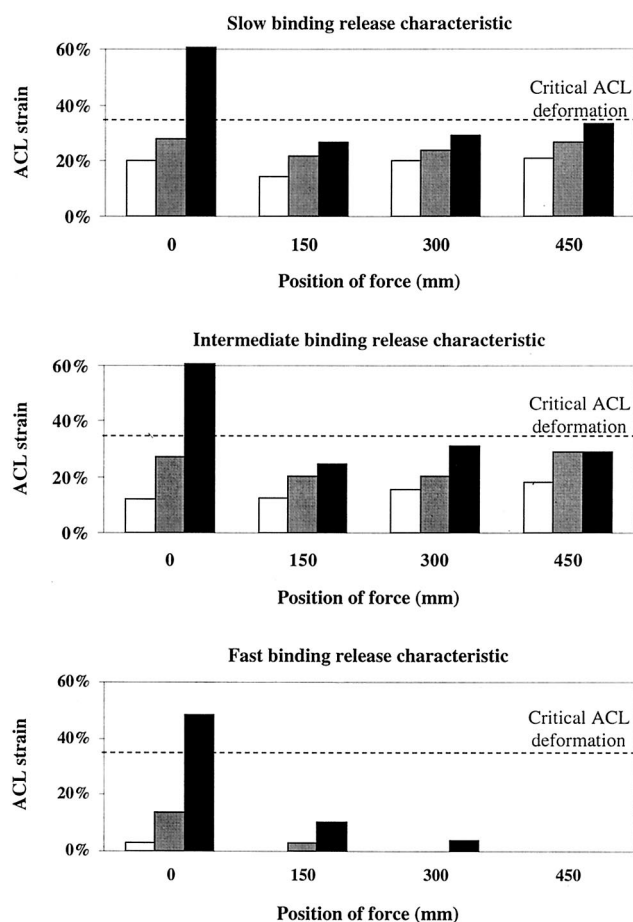


FIGURE 8—Effect of pivot point position on maximal ACL strain. A 600-N load was applied at the projection of the tibial axis as well as 150, 300, or 450 mm behind that point. The pivot was positioned 63 mm in front (white bar), 63 mm in the back (gray bar), or 135 mm in the back (black bar) relative to the center of the boot.

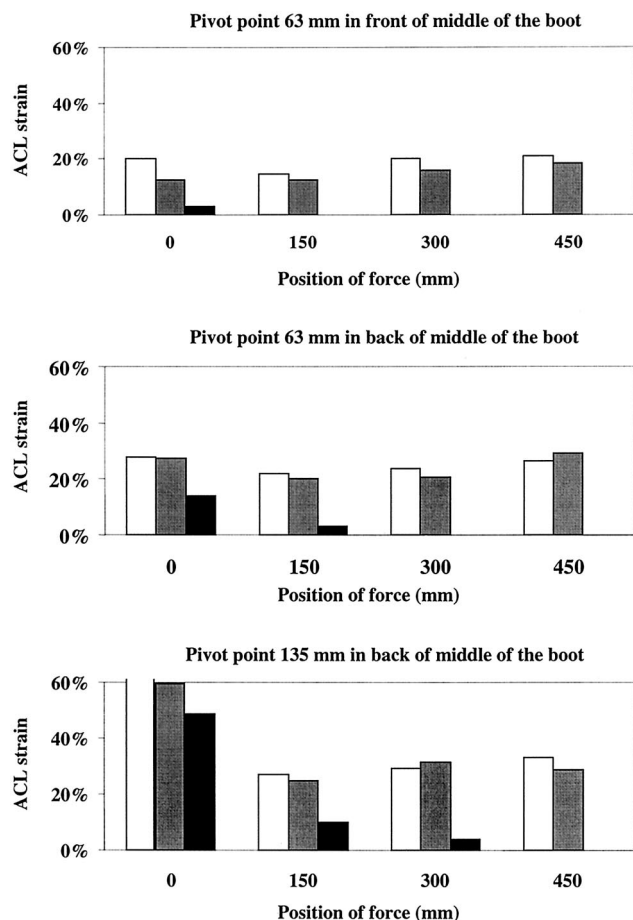


FIGURE 9—Effect of release characteristics on maximal ACL strain. A 600-N load was applied at the projection of the tibial axis as well as 150, 300, or 450 mm behind that point. Slow (white bar), intermediate (gray bar), and fast (black bar) binding release characteristics were simulated.

all three pivot points, ACL strain was less when the fast binding characteristic was simulated, whereas the results were similar for the slow and intermediate characteristics. According to those results, our model predicts, that under simulated phantom-foot conditions, a binding with fast characteristic and a pivot point positioned in front of the center of the boot produces less strain on the ACL. A binding with fast characteristic with a pivot point positioned in front of the center of the boot also produces less strain on the ACL when a load is applied on the medial edge of the ski at the projection of the tibial axis, even though this situation does not reproduce the phantom-foot mechanism. A load applied on the medial edge of the ski at the projection of the tibial axis represents the worst situation, since a traditional binding with a pivot point positioned at the back would not sense it.

We are not suggesting that the pivot point at the back of the binding should be moved in front. We are well aware that some injuries are produced by twisting loads at the tip of the ski. For example, the ski tip might engage the snow or the ski tips could cross. A binding with two pivot points, one positioned in front and one at the back, could however sense twist loads applied to the ski both at the front and at the back, and might therefore be a

solution to reduce the occurrence of knee injuries. Also, having two pivot points on a binding would prevent the blind spot phenomenon. With only one pivot on a binding, there is no torque generated at the pivot point when a load is applied at the position of the pivot. The binding will therefore not release in this situation. With two pivot points on a binding, at least one of them would be effective in every situation.

Critical ACL strain. Ligament strain at maximal load depends on many parameters, such as loading rate and age. Also, under different conditions (magnitude of load, speed of fall), ligament strain might be different. Moreover, due to limitations of the model (see next section), it might not be appropriate to report absolute values. We, therefore, think that it is more appropriate to compare strains generated using different binding release characteristics or different pivot points, rather than compare ligament strain with the maximal strain that can be sustained by the ligament. For example, although ligament strain never reached critical values (35%) when a load was applied behind the projection of the tibial axis, the fact that maximal strain was less for a pivot point positioned in front of the center of the boot means that it is less probable that the ACL will rupture with such a pivot.

Limitations of the model. When building a model to perform a simulation study, it is not possible to deploy everything in detail. Therefore, some aspects have to be neglected, such as muscle activation, contact with snow, detailed characteristics of the binding, etc. In particular, contact between the cruciate ligaments was neglected in the present model. It can thus be assumed that in reality, a knee joint submitted to similar loading conditions might experience greater ACL strains than those predicted by our model. Therefore, we cannot report absolute values but only differences between different situations. It is important to keep in mind that the results obtained are true under simulated phantom-foot conditions and that we have to be careful when extrapolating these results to the ski hill.

In conclusion, computer modeling allows evaluation of many situations that would be impossible in *in vitro* or *in vivo* experiments. Our model predicts that under simulated phantom-foot conditions, a binding with fast characteristics with a pivot point positioned in front of the center of the boot produces less strain on the ACL. A binding with two pivot points, one positioned in front and the other at the back, could sense twist loads applied to the ski both at the front and at the back, and might therefore be a solution to reduce the occurrence of ACL injuries. Future investigations include applying different loads on the ski to reproduce a different injury mechanism. Strain of other ligaments could also be computed.

We are grateful to Rick Howell for sharing his knowledge of ski research. The editorial work of Mr. Ovid Da Silva, Editor/reviewer of the Research Support Office, Research Centre of the University of Montreal Hospital Center, is also appreciated.

The authors acknowledge the financial support provided by Natural Science and Engineering Research Council of Canada (NSERC), Industrial Research Assistance Program (IRAP) and Karhu Canada.

REFERENCES

1. BENJABALLAH, M. Z., A. SHIRAZI-ADL, and D. J. ZUKOR. Biomechanics of the human knee joint in compression : reconstruction, mesh generation and finite element analysis. *The Knee* 2:69–79, 1995.
2. BEYNNON, B. D., R. J. JOHNSON, B. C. FLEMING, C. J. STANKEWICH, P. A. RENSTROM, and C. E. NICHOLS. The strain behavior of the anterior cruciate ligament during squatting and active flexion-extension. A comparison of an open and a closed kinetic chain exercise. *Am. J. Sports Med.* 25:823–829, 1997.
3. BIDEN, E., and J. O'CONNOR. Experimental methods used to evaluate knee ligament function. In: *Knee Ligaments: Structure, Function, Injury, and Repair*. New York: Raven Press, 1990, pp. 135–161.
4. BLANKEVOORT, L., and R. HUISKES. Ligament-bone interaction in a three-dimensional model of the knee. *J. Biomech. Eng.* 113:263–269, 1991.
5. DEIBERT, M. C., D. D. ARONSSON, R. J. JOHNSON, C. F. ETTLINGER, and J. E. SHEALY. Skiing injuries in children, adolescents, and adults. *J. Bone Joint Surg. (Am.)* 80:25–32, 1998.
6. EKELAND, A., and B. O. THORESEN. Isolated rupture of the anterior cruciate ligament by knee hyperflexion. In: *Skiing Trauma and Safety: Sixth International Symposium*, C. D. J. Mote and R. J. Johnson (Eds.). Philadelphia: American Society for Testing and Materials, 1987, pp. 61–67.
7. ETTLINGER, C. F., and R. J. JOHNSON. Can knee injuries be prevented? *Skiing* 43:120–123, 1991.
8. ETTLINGER, C. F., R. J. JOHNSON, and J. E. SHEALY. A method to help reduce the risk of serious knee sprains incurred in alpine skiing. *Am. J. Sports Med.* 23:531–537, 1995.
9. FISCHER, J. F., P. F. LEYVRAS, and A. BALLY. A dynamic analysis of knee ligament injuries in alpine skiing. *Acta Orthop. Belg.* 60:194–203, 1994.
10. FLEMING, B. C., P. A. RENSTROM, B. D. BEYNNON, et al. The effect of weightbearing and external loading on anterior cruciate ligament strain. *J. Biomech.* 34:163–170, 2001.
11. GROOM, E. S., and W. J. SUNTAY. A joint coordinate system for the clinical description of three-dimensional motions: application to the knee. *J. Biomech. Eng.* 105:136–144, 1983.
12. HAGEMEISTER, N. *In vitro Evaluation of Different Surgical Reconstructions of Cruciate Ligaments of the Knee*. Ph.D. Thesis, Ecole Polytechnique de Montreal, Universite of Montreal, Montreal, Canada, 2001, pp. 1–243.
13. HAGEMEISTER, N., R. LONG, L. YAHIA, et al. Quantitative comparison of three different types of anterior cruciate ligament reconstruction methods: laxity and 3-D kinematic measurements. *Biomed. Materials Eng.* 12:47–57, 2002.
14. HIROKAWA, S., M. SOLOMONOW, Y. LU, Z. P. LOU, and R. D'AMBROSIA. Anterior-posterior and rotational displacement of the tibia elicited by quadriceps contraction. *Am. J. Sports Med.* 20: 299–306, 1992.
15. JOHNSON, R. J., and P. A. RENSTROM. Injuries in alpine skiing. In: *Clinical Practice of Sports Injury, Prevention and Care*, P. A. Renstrom (Ed.). Boston: Blackwell Scientific Publications, 1994, pp. 676–698.
16. JOHNSON, R. J., C. F. ETTLINGER, R. J. CAMPBELL, and M. H. POPE. Trends in skiing injuries: analysis of a 6-year study (1972 to 1978). *Am. J. Sports Med.* 8:106–113, 1980.
17. JOHNSON, R. J., C. ETTLINGER, and J. E. SHEALY. Skier injury trends. In: *Skiing Trauma and Safety: Seventh International Symposium*, R. J. Johnson, C. D. J. Mote, and M.-H. Binet (Eds.). Philadelphia: American Society for Testing and Materials, 1989, pp. 25–31.
18. JOHNSON, R. J., C. ETTLINGER, and J. E. SHEALY. Update on injury trends in alpine skiing. In: *Skiing Trauma and Safety*, Vol. 13, R. J. Johnson, P. Zucco, and J. E. Shealy (Eds.). West Conshohocken: American Society for Testing and Materials, 2000, pp. 108–118.
19. KENNEDY, J. C., R. J. HAWKINS, R. B. WILLIS, and K. D. DANYLCHUCK. Tension studies of human knee ligaments. yield point, ultimate failure, and disruption of the cruciate and tibial collateral ligaments. *J. Bone Joint Surg. (Am.)* 58:350–355, 1976.
20. LAMONT, M. K. New Zealand ski injury statistics: 1989 and 1990 ski seasons. In: *Skiing Trauma and Safety: Ninth International Symposium*, R. J. Johnson, C. D. J. Mote, and J. Zelcer (Eds.). Philadelphia: American Society for Testing and Materials, 1993, pp. 33–42.
21. MARIEB, E. N., and G. LAURENDEAU. Les articulations. In: *Anatomie et physiologie humaines*. Éditions du renouveau pédagogique inc. Canada: Ville St-Laurent, 1993, pp. 222–245.
22. MARINOZZI, G., S. PAPPALARDO, and R. STEINDLER. Human knee ligaments: mechanical tests and ultrastructural observations. *Ital. J. Orthop. Traumatol.* 9:231–240, 1983.
23. MARSHALL, J. R., R. WARREN, and D. J. FLEISS. Ligamentous injuries of the knee in skiing. *Clin. Orthop. Relat. Res.* 108:196–199, 1975.
24. MOW, V., G. ATESHIAN, and R. L. SPILKER. Biomechanics of diarthrodial joints: a review of twenty years of progress. *J. Biomech. Eng.* 115:460–467, 1993.
25. NATRI, A., B. D. BEYNNON, C. F. ETTLINGER, R. J. JOHNSON, and J. E. SHEALY. Alpine ski bindings and injuries: current findings. *Sports Med.* 28:35–48, 1999.
26. PIAZZA, S. J., and S. L. DELP. Three-dimensional dynamic simulation of total knee replacement motion during a step-up task. *J. Biomech. Eng.* 123:599–606, 2001.
27. SHEALY, J. E. Overall analysis of NSAA/ASTM data on skiing injuries for 1978 through 1981. In: *Skiing Trauma and Safety: Fifth International Symposium*, R. J. Johnson and C. D. J. Mote (Eds.). Philadelphia: American Society for Testing and Materials, 1985, pp. 302–313.
28. SHEALY, J. E. Comparison of downhill ski injury patterns: 1978–81 vs. 1988–90. In: *Skiing Trauma and Safety: Ninth International Symposium*, R. J. Johnson, C. D. J. Mote, and J. Zelcer (Eds.). Philadelphia: American Society for Testing and Materials, 1993, pp. 23–32.